JIAHUI HUANG (黄家辉)

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RESEARCH INTERESTS

Multi-Agent Learning · Reinforcement Learning · Generative Models · Agentic AI

EDUCATION

- **M.Sc. in Engineering (Robotics and Intelligent Systems)**, The University of Hong Kong, 2026.9 - 2027.6 (expected)
- **B.Eng. in Artificial Intelligence**, Chongqing University, 2022.9 - 2026.6 · GPA 3.5/4.0
  B.Eng. Thesis: *Design and Implementation of an LLM-based Enterprise Fundamental Analysis Agent*

PUBLICATIONS & MANUSCRIPTS

- **Risk-sensitive reinforcement learning with generative value distributions** [NeurIPS 2026, under review] · first author
  Models the full return distribution via conditional flow matching and jointly trains mean- and CVaR-optimized actors on a shared flow-based critic; competitive reward with 7 baselines and a substantially lower crash rate.
- **Cause-specific local repair for scalable multi-agent path finding** [AAAI 2027, under review] · first author
  Propose a local-repair framework that decomposes the repairable slack of a MAPF solution into goal-lock, detour and precedence causes, with a dedicated repair operator per cause. 100% solve rate on 5,850 MovingAI instances across 10 map types, ~2.0 s average runtime; at a matched 2 s budget, excess sum-of-costs (SoC) is 5.1 percentage points below LaCAM3.
- Three additional first-author manuscripts under review at AAAI 2027: generative experience replay for continual off-policy RL; robot controller interface calibration; distributionally robust learning under calibrated shifts.

RESEARCH EXPERIENCE

- **Algorithm Engineer, Shanghai Xianruan Information Technology** 04/2025 - 06/2025
  - Built the core solver module of a multi-robot warehouse path planning system: a Python ECBS implementation with task allocation, designed for production-scale warehouse deployment.
  - Tuned the conflict-detection mechanism and heuristics to product-scale instances: +40% solve speed, +30% throughput, under high-concurrency workloads.

RESEARCH PROJECTS

Agentic AI

- **Vellorys** — an LLM-based multi-agent research system for equity analysis across CN/HK/US/crypto markets. Designed a LangGraph-based agent workflow with upfront tool evidence registration, formal argumentation for conflict resolution, and a PPO router for adaptive route selection. 1.9% hallucination rate, 0.960 fact accuracy on 300 FinDebate-Eval samples.
  vellorys.com

Multi-Agent Path Finding

- **GCSS** — a guided large-scale MAPF solver: scatter-guided multi-strategy PIBT with automatic swap handling; decentralized priority-based execution that scales to 500-1000 agents in sub-second to seconds. 87.5% solve rate across 33 MovingAI official maps, SoC within 2% of LaCAM3.
- **DiffRefine** — learning-based MAPF trajectory refinement: SE(2)-equivariant GNN plus Mamba sequence model, B-spline parameterization; complete training/eval pipeline with test suite.
- Classical implementations: A*, Focal A*, SIPP, CBS in Python and C++.

Robot Learning

- **VLA: Rectified Flow Vision-Language-Action** — one-step generation at 170 Hz; in our evaluation, 1-step MSE is 98.3% lower than DDPM.
- **Robotic deployments** — navigation and control on Scout Mini, camera-IMU calibration (Kalibr, 1.10 px), LiDAR odometry (NDT/ICP/GN-ICP). Planning-control · Camera-calibration · lidar

SKILLS

Programming: Python, C++, Rust, Go, SQL, LaTeX
ML / RL: PyTorch, SAC, PPO, DQN, distributional RL, diffusion and flow matching, GNNs, Mamba
Multi-Agent / Planning: MAPF, A*, CBS, ECBS, SIPP, PIBT
Robotics: ROS, MuJoCo, Gazebo, PCL, Kalibr